

به نام پروردگار هستی بخش
 دانشگاه علوم پزشکی و خدمات بهداشتی - درمانی قزوین
 دانشکده پزشکی
 نیمسال : اول 03-04



نام درس: بیوشیمی بالینی	نام استاد: دکتر مجتبی فتحی	تعداد سوالات : 39	زمان: دقیقه	تاریخ امتحان:
نام و نام خانوادگی :	شماره دانشجویی: پاسخ سوالات در پاسخنامه وارد گردد.	نمره منفی: ندارد		

بیوشیمی بالینی	
1	C- پپتید در کدام یک از حالات زیر کمتر از بقیه است
الف	انسولینوما
ب	دیابت تایپ I
	هایپر انسولینمیا
د	دیابت دوره بارداری
منبع:	

بیوشیمی بالینی	
2	بیماری دیابت ایجاد شده در اثر مصرف Diazoxide جزو کدام دسته از دیابت محسوب می شود.
الف	نوع 1
ب	نوع 2
ج	دیابت بارداری
	انواع دیگر دیابت
منبع:	

بیوشیمی بالینی	
3	تعداد پیوند های دی سولفیدی در ساختار انسولین چند عدد است
الف	1
ب	2
ج	4
د	0
منبع:	

1. In which of the following conditions is C-peptide lowest (lower than the rest)?

- a. Insulinoma
- b. Type I diabetes ✓
- c. Hyperinsulinemia
- d. Gestational diabetes

2. Diabetes caused by the use of Diazoxide belongs to which category of diabetes?

- a. Type 1
- b. Type 2
- c. Gestational diabetes
- d. Other specific types of diabetes ✓

3. How many disulfide bonds are in the structure of insulin?

- a. 1
- b. 2 ✓
- c. 4
- d. 0

4. A person with features of Reaven syndrome (metabolic syndrome) is at high risk for which category of disease?

- a. Cancer
- b. Osteoporosis
- c. Cardiac (cardiovascular) ✓
- d. Endocrinopathy

5. In Von Gierke disease, all of the following laboratory findings are seen, EXCEPT:

- a. Lactic acidemia
- b. Hypoglycemia
- c. Hypouricemia ✓
- d. Hepatomegaly

6. Hereditary fructose intolerance and dietary fructose intolerance are respectively due to a defect in which of the following?

- a. Fructokinase – GLUT1
- b. Phosphofructokinase – Fructokinase
- c. Aldolase B – GLUT5 ✓
- d. Aldolase A – GLUT3

7. Cataract and formation of AGE compounds arise from which carbohydrate?

- a. Mannose
- b. Galactose ✓
- c. Sucrose
- d. Fructose

8. All of the following are seen in McArdle disease, EXCEPT:

- a. Hepatomegaly ✓
- b. Muscle cramps
- c. Myoglobinuria
- d. Iron deposition in the kidney and renal failure

9. Which of the following functions is related to farnesyl pyrophosphate?

- a. Formation of HMG-CoA ✓
- b. Formation of vitamin K
- c. Formation of vitamin A
- d. Formation of dolichol phosphate

10. How many units of isopentenyl pyrophosphate are needed to form geraniol pyrophosphate (geranyl-PP)?

- a. 3
- b. 2 ✓
- c. 4
- d. 1

11. Which of the following is a laboratory finding of SLOS (Smith–Lemli–Opitz Syndrome)?

- a. Lymphopenia
- b. Increased lanosterol
- c. Thrombocytosis
- d. Normocytic normochromic anemia ✓

12. Which apolipoprotein plays an essential role in the formation of HDL?

- a. Apo C-II
- b. Apo B100
- c. Apo B48
- d. Apo A-I ✓

13. In which of the Fredrickson classifications is there NO decrease in HDL?

- a. Type I
- b. Type II-a ✓
- c. Type IV
- d. Type V

14. Which antihypertensive drug leads to hyperkalemia?

- a. Acetazolamide
- b. Triamterene ✓
- c. Furosemide
- d. Thiazide

15. In which of the following conditions does blood creatinine show a falsely low value?

- a. High urea
- b. High hemoglobin ✓
- c. High blood glucose
- d. Use of cephalosporins

16. What is the GFR value in ESRD patients?

- a. Less than 15 ✓
- b. 15–30
- c. 30–60
- d. Less than 30

17. For early detection of kidney diseases, measuring which is more appropriate?

- a. Serum creatinine
- b. Cystatin C clearance
- c. Microalbuminuria ✓
- d. Urine osmolarity

18. In which liver disease can the ALT rise exceed 1000 IU/L?

- a. Fatty liver
- b. Liver cirrhosis
- c. Hepatitis C
- d. Hepatitis A ✓

19. Deficiency of which of the following vitamins leads to a falsely low measurement of the enzyme ALT?

- a. B6 ✓
- b. B3
- c. B1
- d. B9

20. Which of the following disorders leads to an increase in direct (conjugated) bilirubin?

- a. Use of contrast media in CT
- b. Dubin–Johnson syndrome ✓
- c. Gilbert syndrome
- d. Hemolytic anemia

21. A person with a marked increase in alkaline phosphatase and gamma-GT probably has which liver disorder?

- a. Gilbert syndrome
- b. Rotor syndrome
- c. Biliary duct obstruction ✓
- d. Fatty liver

22. In which of the following porphyrias do clinical symptoms appear in childhood and with greater severity?

- a. AIP
- b. HCP (Hereditary Coproporphyria)
- c. VP
- d. CEP ✓

23. What is the most common acquired porphyria?

- a. CEP (Günther disease)
- b. Acute Intermittent Porphyria
- c. PCT (Porphyria Cutanea Tarda) ✓
- d. EPP (Erythropoietic Protoporphyria)

24. In PCT porphyria, which compound increases in urine or feces?

- a. PBG in urine
- b. Uroporphyrin in urine or feces ✓
- c. ALA in urine
- d. PBG in feces

25. Which of the heme biosynthesis enzymes is present (also) in the mitochondria?

- a. ALA synthase ✓
- b. ALA dehydratase
- c. PBG deaminase
- d. Uroporphyrinogen cosynthase

27. Which of the following symptoms is NOT present in phenylketonuria (PKU)?

- a. Liver cirrhosis ✓
- b. Self-harm (self-injurious behavior)
- c. Paralysis
- d. Mousy odor of urine and sweat

28. What is the most important clinical sign in alkaptonuria patients?

- a. Ataxia
- b. Anemia
- c. Facial acne
- d. Ochronosis ✓

30. Deficiency of which vitamin is present in Hartnup disease?

- a. B1
- b. B2
- c. B12
- d. B3 ✓

○ (niacin/B3 — since tryptophan malabsorption reduces niacin synthesis)

31. Which of the following amino acids is conditional (conditionally essential)?

- a. Asp
- b. Ala
- c. Ser
- d. Arg ✓

32. In which of the following conditions does the urine NOT develop a caramel/burnt-sugar (maple syrup) smell due to the presence of sotolon?

- a. Tyrosinemia type III ✓
- b. Pregnancy
- c. Prolonged starvation
- d. MSUD disease

33. Which of the following compounds does NOT play a role in the conversion of phenylalanine to tyrosine?

- a. NADPH, H⁺
- b. O₂
- c. BH₄
- d. Cu ✓

34. Formation of cystathionine from homocysteine (homoCys) requires which amino acid?

- a. Alanine
- b. Serine ✓
- c. Leucine
- d. Glycine

35. Which of the following vitamins does NOT play a role in homocysteine (homoCys) metabolism?

- a. B12
- b. B6
- c. B9
- d. B5 ✓

36. Troponin measurement is helpful for diagnosing which patients?

- a. STEMI
- b. Non-STEMI ✓
- c. Pulmonary embolism
- d. Dyspnea (shortness of breath)

37. The "Flipped Ratio" is used for which of the following factors?

- a. ALT
- b. AST
- c. LDH ✓
- d. cTnI

38. Which of the following is more sensitive than the rest for diagnosing myocardial infarction in the early hours?

- a. CK-MB
- b. Myoglobin ✓
- c. LDH
- d. Troponin

39. To differentiate Dyspnea caused by cardiac disease from (that caused by) intravascular coagulation, measuring which is appropriate?

- a. D-dimer ✓
- b. AST
- c. BNP
- d. hsCRP